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Studierichting: Informatica

Oefeningenreeks 1D nr. 20.6

Bepaal de nulpunten en de ontbinding in factoren in $\mathbb{R}[z]$ van:

$$F(x) = z^8 + z^4 + 1$$

Manier 1:

$$\text{stel } t = z^4$$

$$\text{dan } F(x) = t^2 + t + 1$$

$$D = 1 - 4 = -3$$

$$t_1 = \frac{-1 - \sqrt{-3}}{2} = \frac{-1 - i\sqrt{3}}{2} = \text{cis}\left(\frac{-2\pi}{3}\right)$$

$$t_2 = \frac{-1 + \sqrt{-3}}{2} = \frac{-1 + i\sqrt{3}}{2} = \text{cis}\left(\frac{2\pi}{3}\right)$$

$$z_1 = \sqrt[4]{t_1} = \sqrt[4]{1} \text{cis}\left(\frac{\frac{-2\pi}{3} + k2\pi}{4}\right) \text{ met } k \in \{0, 1, 2, 3\}$$

$$(z_1)_1 = \sqrt[4]{1} \text{cis}\left(\frac{\frac{-2\pi}{3}}{4}\right) = \text{cis}\left(\frac{-\pi}{6}\right) = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

$$\Rightarrow (z_2)_1 = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$(z_1)_2 = \sqrt[4]{1} \text{cis}\left(\frac{\frac{-2\pi}{3} + 2\pi}{4}\right) = \text{cis}\left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\Rightarrow (z_2)_2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$(z_1)_3 = \sqrt[4]{1} \text{cis}\left(\frac{\frac{-2\pi}{3} + 4\pi}{4}\right) = \text{cis}\left(\frac{5\pi}{6}\right) = \frac{-\sqrt{3}}{2} + \frac{1}{2}i$$

$$\Rightarrow (z_2)_3 = \frac{-\sqrt{3}}{2} - \frac{1}{2}i$$

$$(z_1)_4 = \sqrt[4]{1} \text{cis}\left(\frac{\frac{-2\pi}{3} + 6\pi}{4}\right) = \text{cis}\left(\frac{4\pi}{3}\right) = \frac{-1}{2} - \frac{\sqrt{3}}{2}i$$

$$\Rightarrow (z_2)_4 = \frac{-1}{2} + \frac{\sqrt{3}}{2}i$$

Er bestaan geen reële nulpunten. Als we elk nulpunt uit z_1 vermenigvuldigen

met zijn complementaire z_2 bekomen we de ontbinding in factoren in $\mathbb{R}[z]$.

$$\begin{aligned} F(x) &= \left[z - \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) \right] \cdot \left[z - \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) \right] \cdot \left[z - \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \right] \cdot \left[z - \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) \right] \\ &\cdot \left[z - \left(\frac{-\sqrt{3}}{2} + \frac{1}{2}i \right) \right] \cdot \left[z - \left(\frac{-\sqrt{3}}{2} - \frac{1}{2}i \right) \right] \cdot \left[z - \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right) \right] \cdot \left[z - \left(\frac{-1}{2} + \frac{\sqrt{3}}{2}i \right) \right] \\ &= (z^2 - \sqrt{3}z + 1) \cdot (z^2 - z + 1) \cdot (z^2 + \sqrt{3}z + 1) \cdot (z^2 + z + 1) \end{aligned}$$

Andere methode:

$$\begin{aligned} F(x) &= z^8 + z^4 + 1 \\ &= z^8 + 2z^4 - z^4 + 1 \\ &= (z^4 + 1)^2 - z^4 \\ &= [(z^4 + 1) - z^2] \cdot [(z^4 + 1) + z^2] \\ &= [z^4 + 1 + 2z^2 - 3z^2] \cdot [z^4 + 1 + 2z^2 - z^2] \\ &= [(z^2 + 1)^2 - 3z^2] \cdot [(z^2 + 1)^2 - z^2] \\ &= [(z^2 + 1) - \sqrt{3}z] \cdot [(z^2 + 1) + \sqrt{3}z] \cdot [(z^2 + 1) - z] \cdot [(z^2 + 1) + z] \\ &= (z^2 - \sqrt{3}z + 1) \cdot (z^2 + \sqrt{3}z + 1) \cdot (z^2 - z + 1) \cdot (z^2 + z + 1) \end{aligned}$$

Ook hieruit kun je afleiden dat er geen reële nulpunten zijn. Je kunt wel 8

nulpunten vinden in $\mathbb{C}$. Dit doen we met de discriminant van de vier

resulterende 2^{de} graads functies.

$$(z^2 - \sqrt{3}z + 1)$$

$$\Delta_1 = (-\sqrt{3})^2 - 4 \cdot 1 \cdot 1 = -1$$

$$\Rightarrow (z_1)_1 = \frac{-(-\sqrt{3}) - \sqrt{-1}}{2 \cdot 1} = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

$$\Rightarrow (z_1)_2 = \frac{-(-\sqrt{3}) + \sqrt{-1}}{2 \cdot 1} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$(z^2 + \sqrt{3}z + 1)$$

$$\Delta_2 = (\sqrt{3})^2 - 4 \cdot 1 \cdot 1 = -1$$

$$\Rightarrow (z_2)_1 = \frac{-\sqrt{3} - \sqrt{-1}}{2 \cdot 1} = \frac{-\sqrt{3}}{2} - \frac{1}{2}i$$

$$\Rightarrow (z_2)_2 = \frac{-\sqrt{3} + \sqrt{-1}}{2 \cdot 1} = \frac{-\sqrt{3}}{2} + \frac{1}{2}i$$

$$(z^2 - z + 1)$$

$$\Delta_3 = (-1)^2 - 4 \cdot 1 \cdot 1 = -3$$

$$\Rightarrow (z_3)_1 = \frac{-(-1) - \sqrt{-3}}{2 \cdot 1} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$\Rightarrow (z_3)_2 = \frac{-(-1) + \sqrt{-3}}{2 \cdot 1} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$(z^2 + z + 1)$$

$$\Delta_4 = 1^2 - 4 \cdot 1 \cdot 1 = -3$$

$$\Rightarrow (z_4)_1 = \frac{-1 - \sqrt{-3}}{2 \cdot 1} = \frac{-1}{2} - \frac{\sqrt{3}}{2}i$$

$$\Rightarrow (z_4)_2 = \frac{-1 + \sqrt{-3}}{2 \cdot 1} = \frac{-1}{2} + \frac{\sqrt{3}}{2}i$$

References